

CHASE S. FOSTER

Data Science | Machine Learning | Statistics

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Portfolio: <https://chasefoster3.github.io/>

EDUCATION

2025 - Present
University of
California, Berkeley

Master of Information and Data Science

- Completed Graduate Coursework: Experimentation and Causal Inference, Applied Machine Learning, Fundamentals of Data Engineering, Research Design, Data Science Programming
- In-Progress Coursework: Natural Language Processing with Deep Learning, Machine Learning Systems Engineering

2021 - 2025
East Tennessee
State University

Bachelor of Science in Mathematics/Statistics

- Graduated Summa Cum Laude
- Major: Mathematics – Statistics Concentration
- Minors: Computer and Information Science and Global Citizenship Honors
- GPA: 3.976

TECHNICAL SKILLS

Programming: Python, R, SQL, Java, C#

Machine Learning and Statistics: Regression, Classification, Clustering, Random Forest, Neural Networks, Dimensionality Reduction, Statistical Inference, Hypothesis Testing, Experimental Design, Power Analysis

Data Science: Exploratory Data Analysis, Data Visualization, Data Cleaning, Feature Engineering

AI/NLP: LLM Evaluation, Embeddings, Semantic Similarity

Libraries & Frameworks: Scikit-learn, TensorFlow/Keras, Pandas, NumPy

Tools: Git, GitHub, Jupyter, RStudio, VSCode, Docker

RELEVANT EXPERIENCE

2023
Kansas City Royals
Dominican Academy

Summer Data Analytics Intern

- Completed an independent research project analyzing Latin American players on MLB 40-man rosters.
- Collaborated directly with baseball operations and video staff, supporting live game and practice data collection in a professional sports environment.
- Gained hands-on experience translating raw performance data into insights.

2022 - 2025
ETSU Center for
Academic
Achievement

Mathematics & Computer Science Tutor

- Delivered individual instruction in probability, statistics, calculus, and computer science to undergraduates across varying skill levels.
- Broke down complex quantitative and computational concepts into accessible explanations, improving student comprehension and performance.
- Built strong technical communication skills through 3 years of sustained one-on-one and group mentorship.

PROJECTS

MLB Postseason Prediction - ETSU - Honors Thesis

- Analyzed 1998–2024 MLB team-season data to identify statistical factors associated with postseason success.
- Evaluated multiple statistical and machine learning approaches, including PCA, clustering, logistic regression, LDA, decision trees, bagging, and Random Forest.
- Identified ERA+, Runs Allowed, OPS+, and Runs Scored among the most important variables for distinguishing postseason outcomes.
- Selected a Random Forest model to generate predictions for the 2025 MLB postseason.

LLM Consistency Experiment - UC Berkeley MIDS - Experimentation & Causal Inference

- Designed and conducted a controlled experiment investigating whether consistency-focused prompting affects the consistency of repeated LLM responses.
- Evaluated Gemini 2.5 Flash, GPT-4.1 Mini, and Gemma-3 across text and image-based tasks using repeated generations under controlled experimental conditions.
- Applied sentence-transformer embeddings (all-mpnet-base-v2) and cosine similarity to quantify semantic consistency across model responses.
- Analyzed treatment effects using regression models, interaction terms, and cluster-robust standard errors to evaluate differences across models and experimental conditions.
- Found that the consistency intervention reduced overall response consistency by approximately 20%, with effects varying substantially across models.

2026 FIFA World Cup Prediction - UC Berkeley MIDS - Applied Machine Learning

- Integrated 49,071 international match records and 13,130 FIFA ranking records, producing a modeling dataset of 25,691 observations.
- Engineered 15 pre-match features to predict international soccer match outcomes while avoiding information unavailable before matches.
- Compared logistic regression, neural networks, and gradient boosting models to evaluate predictive performance.
- Achieved 60.89% test accuracy with logistic regression and used model predictions to simulate the 2026 FIFA World Cup.

NFL Officiating Exploratory Data Analysis - UC Berkeley MIDS - Data Science Programming

- Conducted exploratory analysis of NFL penalties, referee assignments, replay reviews, and game-level data from 2015–2024.
- Integrated multiple datasets while cleaning inconsistent team, referee, date, and game identifiers.
- Investigated trends in penalty frequency, referee behavior, replay reviews, and officiating patterns across seasons and teams.
- Developed visualizations and statistical summaries to communicate trends and identify relationships within the data.

LEADERSHIP

ETSU Army ROTC
2021 - 2022

Contracted Cadet

- Served as team and squad leader during tactical field-training exercises.
- Trained to lead teams and make decisions in simulated high-stress environments.
- Led preparation and execution of morning physical training activities.